

Programare Concurenta si Distribuita


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Week 1

- General presentation
- Course:
 - Distributed Systems. Basic concepts (I).
- Laboratory:
 - Discussions:
 - [Brewer Theorem - CAP](#)
 - [PACELC](#)
 - Shared Memory versus Message Passing
 - Register [here](#).
 - Homework 1. Read [the document](#) for details.

Week 2

- Courses [14:00-20:00]
 - 14:00 - 16:00 - Prerequisites before Cloud Computing
 - 16:00 - 18:00 - Distributed Systems. Basic concepts (II).
 - 18:00 - 20:00 - Distributed systems principles applied in Cloud

Week 2'

- Course[March, 8 - 10:00-12:00]: Cloud Service Providers

Week 3

- Laboratory [14:00 - 20:00]: Discussions over Homework 1. Don't forget to register (use the link from Discord)

Week 3'

- Course[March, 14 - 08:00-10:00]: Virtualization in Cloud

Week 4

- Course:
 - [14:00-16:00] Course: Google Cloud Platform
 - [16:00-18:00] Distributed Systems. Basic concepts (III).
- Laboratory [18-20]: Cloud Services & Various Communication mechanisms
 - [Training Credits | Google for Education](#)
 - [Google Cloud](#)
 - [Google Cloud Platform Tour](#)

- Google App Engine, Google Cloud Storage, Google Prediction APIs.
- NodeJS in GAE (from March 2016)
- IAM - Cloud Identity and Access Management
- Real-Time Web Technologies
- Let's remember!: WebSockets, REST APIs, HATEOAS
- Introducing WebSockets
- What Are Web Sockets?
- Web Socket API
- Web Socket - exemplu
- PHP - websockets
- Jetty
- Cloud Native Computing Foundation
- gRPC
- AWS Well-Architected Tool to review your app architecture
- Homework 2:
 - A) Create a distributed application that uses a cloud ecosystem (e.g. Google, Amazon, et.al.). Requirements - the application has at least three components:
 - one is using at least three native cloud services (one is Stateful)
 - the other can be an on-premise application but is required to use at least one service from Google Cloud (or similar services for other Clouds Providers): Cloud BigTable, BigQuery, Cloud DataProc, or services such as Cloud Endpoints, Prediction API
 - the other is a FaaS (AWS Lambda, Google Cloud Functions, IBM OpenWhisk or Microsoft Azure Functions)
 At least one Real-Time Web Technology (e.g. WebSocket, gRPC) must be used in a proper manner in your system. (Don't forget about performance metrics that fit to your distributed application). More details are offered during the laboratory.
 - B) Create a scientific report based on the services chosen in point A, which are used in various distributed architectures of real systems (e.g. Twitter). Analyze the characteristics of your system: performance, latency, reliability, transparency dimensions et.al (see our courses)

Week 5 [March 25]

- [14:00-16:00]Laboratory: P2P Discussions over Final Projects & Some Homeworks[2]
- Presentation & Hands-on Lab [16:00-20:00]: Containers&Kubernetes
- To be installed before:
 - Docker Desktop - <https://docs.docker.com/desktop/>
 - Kubernetes local, with Kind - <https://kind.sigs.k8s.io/docs/user/quick-start/#installation> (if there are issues with, you can use minikube: <https://minikube.sigs.k8s.io/docs/start/>)
 - kubectl binary: <https://kubernetes.io/docs/tasks/tools/#kubectl>
- Supplementary resources:
 - [Kubernetes overview](#)
 - [AKS](#)

!!! Week 6 [March 31 - April 3, 2025] - SpringSchool - Fil Library - Floor 3

- Faculty of Computer Science organize the Spring School for students on topics of Cloud/Edge Computing, Network Communication Architectures, Virtualisation Technologies, 5G Networks, Security in Distributed Systems
- Notice: The topics addressed in the SpringSchool cover multiple directions addressed in our courses, but some are required and participation in them is considered as the advanced recovery of these thematic.
- Other details regarding participation to these courses are given through your email, after your REGISTRATION
- The REQUIRED courses from this SpringSchool 2025:
 - March 31 - [10:00-11:00] Interactive Education with Cisco NetAcad: IoT Lab in Packet Tracer
 - April 1 - [09:00-10:45] Google Agent Space
 - April 1 - [13:00-14:00] Increase Development Efficiency with Continuous Integration & Testing
 - April 2 - [12:30-16:30] Introductory course on the principles, case studies, and basic methods of system design, with applications in the field of telecommunications
 - April 3 - [15:00-16:30] Edge Computing: Bringing AI Closer
 - BONUS! Participarea activa la activitatile propuse in 31 Martie - 3 Aprilie 2025 poate conduce la un bonus de 1p la nota finala. Nu uitati sa va treceti pe foaia de prezenta care va fi in fiecare dimineata disponibila la intrare.
- Course was recovered in advance.
- Laboratory [14:00 - 16:00]& [18:00 - 20:00]: Discussions over Homework[2].

Week 7

- Course&Laboratory was recovered in advance.

Week 8

- Evaluation - Test 1 .
- Obs. To contribute to [iTransfer](#) center (Faculty of Computer Science Center for Technological Transfer) fill the following [form](#).

Week 9[April 29]

- Course:
 - [14:00-16:00] Will be recover on April 30, 8:00 - 10:00 | C309- on the direction: Make it happen with Google AI
 - [16:00-20:00] Course&Laboratory: [OpenDSU - PrivateSky Platform & Self Sovereign Applications](#)
 - Obs. The homework [3-4] will be hard to solve without attending this part.
 - Homework[3-4]- Create a document containing: a presentation of the chosen project, argue the need to use PrivateSky technologies for the chosen project, briefly describe the design and implementation. Other details will be given during the presentation.
 - Usefull paper: [Sinica Alboaie, Nicu-Cosmin Ursache, Lenuta Alboaie, Self-Sovereign Applications: return control of data back to people, Knowledge-Based and Intelligent Information & Engineering Systems: Proceedings of the 24th International Conference KES2020, vol. 176, Septembrie 2020, doi: <https://doi.org/10.1016/j.procs.2020.09.183>](#)

Week 10[May 6]

- [16:00-18:00] Will be recover on [May 7] Course: [Online][8:00-10:00][Link Discord]: Analysis of the architecture of a Cloud application (using AWS)
- Also remember Week 6 - Course on the principles, case studies, and basic methods of system design, with applications in the field of telecommunications

Week 11[May 13]

- This week was recovered in advance in Week 6

Week 12 [May 20]

- 14:00 - 15:00 - Practical Perspectives on Cloud Application Architecture and Orchestration in AWS
- 15:00 - 16:00 - Individual discussions over final Projects and Homework 5 specifications
- 16:00 - 19:00 - Discussions and evaluations - Homework 3-4.

Week 13 [May,27, 2025]

- Courses [14:00-16:00]:DFS(Distributed File Systems)
- [16:00 - 20:00] Discussions and evaluations - Homework 5 || Projects

Week 14 [Iunie, 3, 2025]

- Courses [14:00-16:00]: GFS&HDFS&Hadoop Ecosystem
- Discussions and evaluations - Homework 5 || Projects

Test 2

[Test 2] June, 11, 2025 - [10:00 - 12:00] - Room C2

- Obs. You can receive questions from this semester's courses and laboratories.
- Obs. Please verify this page for more updates.

Observatii:

Pentru a fi vizualizate optim, prezentarile necesita Adobe Reader (disponibil gratuit la <http://www.adobe.com/acrobat/>).